1 (i) $>$ **[B1]**

(ii) $>$ **[B1]**

[Total Marks: 2]

2 Converts to a common format (e.g. decimals: $0.625, 0.62, 0.615, 0.6$) **[M1]**

$\frac{3}{5}, 61.5\%, 0.62, \frac{5}{8}$ **[A1]**

[Total Marks: 2]

3 (a) $>$ **[B1]**

(b) $=$ **[B1]**

(c) $<$ **[B1]**

[Total Marks: 3]

4 (a) (i) $<$ **[B1]**

(ii) $=$ **[B1]**

(b) M1 for converting at least three values into a comparable form (e.g. percentages: 55%, $54.54\%\ldots$, 54%, $55.55\%\ldots$) **[M1]**

$54\%, \frac{6}{11}, 0.55, \frac{5}{9}$ **[A1]**

[Total Marks: 4]

5 (a) Converting to comparable decimals: 0.75, 0.72, 0.74, $0.714\ldots$, $0.744\ldots$ (at least three correct) **[M1]**

$\frac{5}{7}$, 0.72, 74%, $0.7\dot{4}$, $\frac{3}{4}$ **[A1]**

(b) (i) $>$ **[B1]**

(ii) $=$ **[B1]**

[Total Marks: 4]

6 (a) (i) $=$ **[B1]**

(ii) $>$ **[B1]**

(iii) $>$ **[B1]**

(b) 0.3 33% $\frac{1}{3}$ $\frac{3}{8}$ **[B2]**

If 0 scored, allow [M1] for converting at least two of the values correctly to decimals, e.g. 0.33, $0.333\ldots$, 0.375

[Total Marks: 5]

7 **(a)** $p^2 \quad p \quad \sqrt{p} \quad \frac{1}{p}$ **[B2]**

M1 for evaluating the expressions with a chosen test value for p (e.g., $p = 0.25$) OR for any 3 expressions in the correct relative order.

(b) **(i)** $>$ **[B1]**

(ii) $<$ **[B1]**

[Total Marks: 4]

8 **(a)** $x^2, x, \sqrt{x}, \frac{1}{x}$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

(b) $x^{-2}, \frac{1}{x}, x, x^3$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

[Total Marks: 4]

9 **(a)** M1 for correctly calculating at least two values: $x^2 = 0.16$, $x^3 = -0.064$, $\frac{1}{x} = -2.5$ **[M1]**

$\frac{1}{x}, x, x^3, x^2$ **[A1]**

(b) M1 for squaring both terms to get 75 and 80, OR expressing as $\sqrt{75}$ and $\sqrt{80}$ **[M1]**

$<$ **[A1]**

(c) M1 for identifying a common denominator, e.g. cross multiplying to get $9m < 35$ **[M1]**

M1 for $35 < 9(m + 1)$ leading to $26 < 9m$ **[M1]**

3 **[A1]**

[Total Marks: 7]

10 **(a)** $p^2 \quad p \quad \sqrt{p} \quad \frac{1}{p}$ **[B2]**

If 0 scored, allow [M1] for a correct order of any three of the terms or for successfully substituting a test value such as $p = \frac{1}{4}$ to evaluate all four expressions.

(b) **(i)** $<$ **[B1]**

(ii) $<$ **[B1]**

(c) c, b, a or $5^{20}, 3^{30}, 2^{50}$ **[A1]**

Shows index manipulation to obtain a common power, e.g. $(2^5)^{10}$, $(3^3)^{10}$ and $(5^2)^{10}$ **[M1]**

Evaluates the bases correctly as 32, 27 and 25 **[M1]**

[Total Marks: 7]

11 **(a)** Testing a valid value such as $x = \frac{1}{4}$ to obtain $\frac{1}{16},\ \frac{1}{2},\ 4,\ \frac{1}{4}$ or comparing powers of x **[M1]**

$x^2,\ x,\ \sqrt{x},\ \frac{1}{x}$ **[A1]**

(b) Substituting a valid test value such as $k = -2$ to get $-8,\ -\frac{1}{2},\ 4,\ -\frac{1}{4}$ **[M1]**

Correct ordering of three negative terms: $k^3 < \frac{1}{k} < -\frac{1}{k^2}$ **[M1]**

$k^3,\ \frac{1}{k},\ -\frac{1}{k^2},\ k^2$ **[A1]**

(c) Rearranging to $n^2 - n \leq 0$ or $n(n-1) \leq 0$ **[M1]**

Identifying critical values $n = 0$ and $n = 1$ **[M1]**

$0 < n \leq 1$ **[A1]**

[Total Marks: 8]

12 $0.12 \qquad 12.1\% \qquad \frac{4}{33} \qquad \sqrt{0.015}$ **[B2]**

M1 for correct decimal equivalents seen for at least two of the non-decimal values (e.g., 0.121, 0.1212..., 0.1224...).

[Total Marks: 2]

13 Converts all to a common unit, e.g. grams: 0.52 g, 0.04 g, 4.5 g, 0.6 g **[M2]**

(M1 for 2 correct conversions to a common unit)

4.5×10^3 mg, 6×10^{-7} tonnes, 5.2×10^{-4} kg, 0.04 g **[A1]**

[Total Marks: 3]

14 **(a)** B1 for at least two correct evaluated values ($a = 3100$, $b = 3130.49\ldots$, $c = 3125$, $d = 3136$) **[B1]**

B1 for all four evaluated correctly **[B1]**

d, b, c, a **[A1]**

(b) 15 **[B1]**

[Total Marks: 4]

15 **(a)** $5^{-3} \qquad 0.00805 \qquad \frac{1}{120} \qquad 8.5 \times 10^{-3}$ **[B2]**

If 0 scored, allow [M1] for evaluating at least two correctly as standard decimals: 0.008, 0.0085, 0.00833...

(b) $11.5 \leq m < 12.5$ **[B2]**

If 0 scored, allow [M1] for identifying the bounds 11.5 and 12.5

[Total Marks: 4]

16 (a) Converts at least two values to decimals or percentages correctly (e.g. 0.875, 0.86, 0.833...) **[M1]**

$\frac{5}{6}$, 86%, 0.865, $\frac{7}{8}$ **[A1]**

(b) Finds a common denominator (e.g. 6, 12, etc.) **[M1]**

Any correct fraction (e.g. $\frac{5}{12}$, $\frac{2}{5}$) **[A1]**

[Total Marks: 4]

17 (a) Evaluating decimals: $p = 0.0048$, $q \approx 0.004878$, $r \approx 0.004796$, $s = 0.00485$, $t = 0.0049$ (at least three correct) **[M1]**

Correct ordering of decimal values **[M1]**

t, q, s, p, r (accept original numerical expressions) **[A1]**

(b) Any correct value in standard form strictly greater than $4.7958\cdots \times 10^{-3}$ and less than or equal to 4.8×10^{-3} (e.g. 4.798×10^{-3} or 4.8×10^{-3}) **[B1]**

(c) Always true **[B1]**

Since $q \approx 0.004878$ and $s = 0.00485$, $q > s$; therefore any $y \geq q$ must also be strictly greater than s **[R1]**

[Total Marks: 6]

18 (a) $<$ **[B1]**

(b) Evaluates at least two expressions correctly (0.25, 0.707..., 2) **[M1]**

a^2, a, $\sqrt{a}$, $\frac{1}{a}$ **[A1]**

(c) False **[B1]**

Provides a valid counterexample, e.g. let $p = 2$ and $q = -3$. Then $2 > -3$, but $\frac{1}{2} > -\frac{1}{3}$. **[B1]**

[Total Marks: 5]